

Часть 1 - минимальная корневая ФС. Пишу свой процесс `init`: простая программа, печатает строку и завершается.

```
cat init.c
```

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18$ cat init.c
#include <stdio.h>
#include <unistd.h>
int main(void){
    printf("hello from my init (pid 1)\n");
    sleep(2);
    return 0;
}
```

Компилирую `init` кросс-компилятором статически.

```
arm-linux-gnueabi-gcc -static -o rootfs-min/init init.c && file rootfs-min/init
```

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18$ arm-linux-gnueabi-gcc -static -o rootfs-min/init init.c && file rootfs-min/init
rootfs-min/init: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1]=d64a9ccad427241c42d6107687a133d89808a5a2, for GNU/Linux 3.2.0, not stripped
```

Упаковываю единственный файл `init` в `initramfs` (`cpio` + `gzip`).

```
cd rootfs-min && echo init | cpio -o -H newc | gzip > ../initramfs-min.cpio.gz
&& ls -lh ../initramfs-min.cpio.gz
```

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18$ cd rootfs-min && echo init | cpio -o -H newc | gzip > ../initramfs-min.cpio.gz && ls -lh ../initramfs-min.cpio.gz
-rw-rw-r-- 1 nm nm 221K июн 17 21:56 initramfs-min.cpio.gz
```

Запускаю в QEMU: ядро находит `init`, он отрабатывает и выходит - ядро ловит панику (`init` завершаться нельзя).

```
qemu-system-arm -M vexpress-a9 -nographic -kernel zImage -dtb $DTB -initrd
initramfs-min.cpio.gz -append "console=ttyAMA0"
```

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18$ qemu-system-arm -M vexpress-a9 -nographic -kernel zImage -dtb $DTB -initrd initramfs-min.cpio.gz -append "console=ttyAMA0"
[ 2.583423] Run /init as init process
hello from my init (pid 1)
[ 4.636063] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
[ 4.637315] CPU: 0 UID: 0 PID: 1 Comm: init Not tainted 6.17.0 #1 NONE
[ 4.637587] Hardware name: ARM-Versatile Express
[ 4.637854] Call trace:
[ 4.638395] unwind_backtrace from show_stack+0x10/0x14
```

Часть 2 - `rootfs` на `BusyBox`. Скачиваю исходники.

```
wget https://busybox.net/downloads/busybox-1.36.1.tar.bz2 && tar xf busybox-1.36.1.tar.bz2
```

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18$ wget https://busybox.net/downloads/busybox-1.36.1.tar.bz2
2250K ..... 93% 127M 0s
2300K ..... 95% 1,01M 0s
2350K ..... 97% 16,6M 0s
2400K ..... 99% 25,9M 0s
2450K ..... 100% 371K=1,4s

2026-06-17 21:56:33 (1,77 MB/s) - '/tmp/bb.tar.bz2' сохранён [2525473/2525473]
```

Конфигурирую: make defconfig + включаю статическую сборку.

`make defconfig && sed -i \x27s/# CONFIG_STATIC is not set/CONFIG_STATIC=y/\x27 .config`

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18/busybox-1.36.1$ make defconfig && sed -i \x27s/# CONFIG_STATIC is not set/CONFIG_STATIC=y/\x27 .config && grep CONFIG_STATIC= .config
Circular buffer size in Kbytes (minimum 4KB) (FEATURE_IPC_SYSLOG_BUFFER_SIZE) [16] (NEW) 16
Linux kernel printk buffer support (FEATURE_KMSG_SYSLOG) [Y/n/?] (NEW) y
CONFIG_STATIC=y
```

Собираю BusyBox кросс-компилятором - один статический бинарник со всеми утилитами.

`make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- && file busybox`

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18/busybox-1.36.1$ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- && file busybox
236 | struct rtattr *tb[TCA_CBQ_MAX+1];
|
make[1]: *** [scripts/Makefile.build:198: networking/tc.o] Ошибка 1
make: *** [Makefile:744: networking] Ошибка 2
busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1]=02a7c810d76411718e38789d1681aef266ac5dca, for GNU/Linux 3.2.0, stripped
```

make install создаёт _install (корневую ФС), упаковываю в initramfs.

`make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- install && (cd _install && find . | cpio -o -H newc | gzip > ../../initramfs-busybox.cpio.gz)`

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18/busybox-1.36.1$ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- install && (cd _install && find . | cpio -o -H newc | gzip > ../../initramfs-busybox.cpio.gz)
make[1]: *** [scripts/Makefile.build:198: networking/tc.o] Ошибка 1
make: *** [Makefile:744: networking] Ошибка 2
-rw-rw-r-- 1 nm nm 1,1M июн 17 21:57 /home/nm/study/eltex/labs/lab-18/initramfs-busybox.cpio.gz
```

Запускаю с `rdinit=/bin/sh` - попадаю в командный интерпретатор, ввожу `uname`, `ls`, `ps` (`ps` пуст - нет `/proc`).

`qemu-system-arm -M vexpress-a9 -nographic -kernel zImage -dtb $DTB -initrd initramfs-busybox.cpio.gz -append "console=ttyAMA0 rdinit=/bin/sh"`

```
nm@nm-Aspire-AL15-42P:~/study/eltex/labs/lab-18$ qemu-system-arm -M vexpress-a9 -nographic -kernel zImage -dtb $DTB -initrd initramfs-busybox.cpio.gz -append "console=ttyAMA0 rdinit=/bin/sh"
[ 2.716394] Run /bin/sh as init process
/bin/sh: can't access tty: job control turned off
uname -a
Linux (none) 6.17.0 #1 SMP Tue Jun 16 19:47:32 +07 2026 armv7l GNU/Linux
~ # ls /bin
arch          dumpkmap      kill          netstat       setpriv
ash           echo          link          nice          setserial
base32        ed            linux32      pidof         sh
base64        egrep         linux64      ping          sleep
busybox       false         ln            ping6         stat
cat           fatattr       login         pipe progress stty
chatr         fdflush       ls            printenv      su
chgrp         fgrep         lsattr       ps            sync
chmod         fsync         lzop         pwd           tar
chown         getopt        makemime     reformime     touch
conspy        grep          mkdir         resume        true
cp            gunzip        mkknod       rev           umount
cpio          gzip          mktemp       rm            uname
cttyhack      hostname     more          rmdir         usleep
date          hush         mount         rpm           vi
dd            ionice        mountpoint   run-parts     watch
df            iostat        mpstat       scriptreplay  zcat
dmesg         ipcalc       mt           sed
dnssdomainname kbd_mode     mv           setarch

~ # ps
PID  USER  TIME  COMMAND
~ # poweroff -f
```